



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

B. Sc. CHEMISTRY

Programme	B.Sc Chemistry				
Course Title	PHYSICAL CHEMISTRY – I: STATES OF MATTER				
Type of Course	MAJOR/MINOR				
Semester	II				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	NCERT or equivalent chemistry syllabus of XI and XII, https://onlinecourses.swayam2.ac.in/nce24_sc07/preview				
Course Summary	Atoms and molecules form the matter that is recognisable for us in the real world, as gases, liquids and solids. Why would they exist as they are? And why would they behave as they do? This course is designed to introduce first year UG students, the physical chemistry of matter in different states of its existence through theory and laboratory experiments. The course explains the various types of interactions between atoms and molecules and their important role in physical and chemical characteristics of the different states of matter. The course introduces the theory and experimental methods that are commonly used to study the various states of matter.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To understand the basic nature of real gases and understand interactions at molecular levels	U	C	Assignments/Quiz designed by the instructor
CO2	To recognise the significance of various interactions in condensed matter	U	C	Assignments/Quiz designed by the instructor

CO3	To analyse the physical properties of liquids through theory and practical experiments	An	P	Seminars and exams
CO4	To explain the regular, periodic arrangement of atoms in solids and appreciate the concept of unit cells	An	P	Seminars/ exams
CO 5	To evaluate and understand the importance of the X-ray diffraction technique for characterisation of crystalline solids	Ap	P	Lab/Discussion/Assignments
CO 6	To execute experiments to determine and tune the various colligative properties of dilute solutions	C	P	Lab/Viva voce exams
* - Remember (R), Understand (U), Apply (Ap), Analyze (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45+30)	Marks
I		GASEOUS STATE	15	33
	1	Kinetic theory of gasses: derivation	1	
	2	Maxwell-Boltzmann distribution of molecular velocities -- Average velocity, RMS velocity and most probable velocity (derivations not required)	2	
	3	Collision theory – Collision diameter- Collision number-Collision frequency - Mean free path – Molecular beams (Mention only)	2	
	4	Real gas- Deviation from ideal behavior- Compressibility factor – Virial equation and Virial coefficients- van der Waals equation of state (derivation required)-features of van der Waals equation - Expression of van der Waals equation in virial form and calculation of Boyle temperature - PV isotherms of real gasses – Andrews' experiments - Continuity of states - Isotherm of van der Waals equation	6	

	5	Critical phenomena - Critical constants - Relationship between critical constants and van der Waals constants - Experimental determination of critical constants - Supercritical carbon dioxide and its applications.	4	
II	LIQUID STATE		8	17
	6	Discussion of different types (with suitable examples) of molecular interactions- dipole-dipole, dipole-induced dipole, induced dipole-induced dipole interactions, Lennard-Jones 6-12 potential.	2	
	7	Properties of liquids- Vapour pressure, Refractive index, Surface tension- Interfacial tension and viscosity - Poiseuille's equation – Explanation of these properties on the basis of intermolecular forces.	3	
	8	Hydrogen bonding in water and other polar molecules, its relevance in biological systems.	2	
	9	Liquids on solid surfaces- Hydrophobic and Hydrophilic, Superhydrophilic and Superhydrophobic surfaces- simple explanation by using the water drop contact angles on surfaces	1	
III	SOLID STATE		15	33
	10	Crystalline and amorphous solids- atomic and molecular solids- nucleation and growth of crystals.	2	
	11	Crystalline Materials – Periodicity- Types of Close packing and packing fraction.	1	
	12	Space Lattice - Unit cell (use models)- Lattice planes and Miller indices (use models) - 7 crystal systems- 14 Bravais lattices- Types of cubic crystals and their planes- Distance formula for cubic systems- Calculation of crystal density (Use of software like Crystal viewer is recommended).	4	
	13	X-ray diffraction- Bragg's law (derivation)- Powder and single crystal X-ray diffraction methods, Atomic scattering factor, Structure factor,	3	
	14	Systematic absences for simple, face centered, and body centered cubic crystals, Analysis of XRD patterns of NaCl, KCl and CsCl. Basic idea of electron and neutron diffraction.	3	
	15	Structural transitions in TiO ₂ - anatase, rutile and brookite phases	1	
	16	Concepts of melting point/boiling point and molecular/atomic/ionic interactions, Examples: CO ₂ , N ₂ , H ₂ O, NH ₃ , NaCl, TiO ₂	1	
IV	SOLUTIONS		7	15

	17	Solubility of gases in liquids – Henry's law and its applications	1	
	18	Colligative properties - Relative lowering of vapour pressure	1	
	19	Colligative properties- Elevation in boiling point and depression in freezing point	1	
	20	Colligative properties- Osmotic pressure - Laws of osmotic pressure - Reverse osmosis and its technological relevance	1	
	21	Determination of molecular mass using colligative properties	1	
	22	Solid Solutions: Substitutional and interstitial solid solutions, Differences between Alloys, Mixtures and Composites. Colloids: Dispersed phase and dispersing medium, Sol, Emulsion, Foam, and Aerosol, Tyndall effect, Nephelometry	2	
V	PHYSICAL CHEMISTRY PRACTICALS		30	
		A minimum of 5 practical experiments out of which ONE EACH from sections 1, 2 and THREE from section 3 must be performed and reported. For plots/graphs, suitable softwares may be used and printed hard copies may be presented. Practical records may be in handwritten or computer-typed printed form. Section 1 1. Determination of cryoscopic constant (K_f) of solid solvent using a solute of known molecular mass. (Solvent: Naphthalene, biphenyl Solute: Naphthalene, biphenyl, 1,4-dichlorobenzene, diphenylamine) 2. Determination of molecular mass of the solute using a solvent of known cryoscopic constant (K_f). (Solvent: Naphthalene, biphenyl Solute: Naphthalene, biphenyl, 1,4-dichlorobenzene, diphenylamine) Section 2 3. Determination of molal transition point depression constant (K_t) of salt hydrate using solute of known molecular mass. (Salt hydrates: $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$, $\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$. Solutes: Urea, Glucose) 4. Determination of molecular mass of the solute using a solvent of known molal transition point depression constant (K_t). (Salt hydrates: $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$, $\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$. Solutes: Urea, Glucose)	3 3 3 3	

		Section 3 5. Determination of viscosity of various liquids using Ostwald's viscometer. 6. Study of glycerine-water system and determination of percentage of glycerine using viscometer [plot composition (c) <i>versus</i> time of flow x density of the solution (td)]. 7. Determination of the surface tension of a liquid or a dilute solution (NaCl / surfactant) using a stalagmometer (drop number method). 8. Determination of composition of glycerine-water mixture by refractive index method. 9. Determination of refractive indices of KCl solutions of different concentrations and unknown concentration of KCl solution. 10. Indexing powder XRD patterns and determination of unit cell parameters of simple and/or bcc and/or fcc systems (Instructors must provide the powder XRD patterns and ask students to index it and calculate unit cell parameters)	3 3 3 3 3 3	
		References: Module I o IV 1. Physical Chemistry: Thermodynamics, Structure and Change, 10th Edition, P. Atkins and J. de Paula, (W. H Freeman and Company, New York) 2. D. A. McQuarrie, J. D. Simon, Physical Chemistry – A Molecular Approach, (Viva, 2001.) 3. Solid State Chemistry and its Applications, 2nd Edition, A R West, (Wiley, 2014) Module V 4. Findlay's Practical Physical Chemistry, Ninth Edition, Revised and Edited by B P Levitt, (Longman, London, 1973) 5. J. B. Yadav, Advanced Practical Physical Chemistry, Goel Publications, Meerut, 2008.. 6. R. C. Das, B. Behra, Experiments in Physical Chemistry, Tata McGraw Hill, New Delhi, 1983.		

		Further reading 7. 1. B. R. Puri, L. R. Sharma, M. S. Pathania, Principles of Physical Chemistry, 46th Edn., Vishal Publishing Company, New Delhi, 2013. 8. G. M. Barrow, Physical Chemistry, 5th Edn., Tata McGraw Hill Education, New Delhi, 2006. 9. F. Daniels, R. A. Alberty, Physical Chemistry, 5th Edn., John Wiley and Sons, Canada, 1980. 10. D. P. Shoemaker, C. W. Garland, Experiments in Physical Chemistry, McGraw-Hill Book Company, New York, 1962. 11. W. G. Palmer, Experimental Physical Chemistry, Cambridge University Press, Cambridge, 2009		
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Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O 5	PS O 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	-	-	3	2	3	2	2	-	2	-	1
CO 2	3	2	-	-	3	2	3	2	1	-	2	-	1
CO 3	3	2	-	-	3	2	3	2	1	-	2	-	1
CO 4	3	2	-	-	3	3	3	2	1	-	1	-	1
CO 5	3	2	2	1	3	3	3	2	1	-	3	-	1
CO 6	2	-	3	3	3	3	3	2	1	2	3	2	1

Correlation Levels:

Level	Correlation
-	Nil

1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment /viva	Practical skill evaluation	End Semester Examinations
CO 1		✓		✓
CO 2		✓		✓
CO 3	✓			✓
CO 4	✓			✓
CO 5	✓	✓		✓
CO 6	✓	✓	✓	✓